# CARLYN LEE

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Accomplished Software Engineer specializing in scalable solutions, system reproducibility, and advanced visualization tools for high-performance computing. Experienced in interdisciplinary collaboration, complex algorithms, and system optimization with: *C/C++, Python, Java, MATLAB, Ansys STK*

## EXPERIENCE

**Oligo Space** Feb 2025 - Present
*Software Engineering*

- Leading the development of software tools leveraging advancements in AI to accelerate design and manufacturing processes of bespoke satellite buses based on customer payloads. Architecting end-to-end spacecraft design automation framework connecting payload requirements to complete satellite configurations.
- Developed and prototyped automated pipelines for S- & X-band link budget analysis, regulatory filings (FCC Part 25 / ITU), and integration with commercial ground-station networks.
- Integrated onboard telecommunications and networking components for flight-software interfaces.
- Contributed to engineering team growth through candidate evaluation, technical assessment frameworks, and establishing onboarding programs with documentation standards and structured mentorship to accelerate new hire productivity.

**NASA Jet Propulsion Laboratory, California Institute of Technology** Aug 2012 - Nov 2024
*Applications Software Engineer* *Pasadena, CA*

- **Tools Development:** Built deep space mission telecom simulation and modeling tools. Led and mentored remote interns on mission-critical systems. Automated operations enabling three Deep Space Network sites to manage the network during day shifts instead of 24/7.
- **Mission Operations:** As the Perseverance Rover telecom chair, responsible for rover communications health using data from Mars. Delivered tools reducing operation response-time from several hours to 20 minutes.
- **Supercomputing:** Developed Lunar terrain mask algorithms for Shaheen II supercomputer, enabling 10m-resolution link coverage maps. Parallelized Deep Space Optical Communications simulations, achieving bit error rate of 10e-8, a 1000x improvement.
- **Fieldwork:** Developed reliable multi-agent data transfer processes at transport layer. Field tested autonomous systems in maritime and subterranean environments, contributing to 1st place in DARPA's Urban Circuit Challenge.
- **Verification & Validation:** Developed automated tools and test procedures for avionics flight software, executing tests on flight hardware.

## EXTRACURRICULAR, VOLUNTEER & PROFESSIONAL AFFILIATIONS

**2019 - Present** Global Network Advancement (GNA-G) Data Intensive Sciences Working Group, enabling high-throughput data transfers for Caltech, SCinet demonstration at Supercomputing 24 & 25.
**2018 - Present** SoCal Linux Expo Volunteer, deploying on-site network infrastructure and AV systems.
**2014 - Present** Interplanetary Small Satellite Conference committee member, managing budgets and logistics for 100+ industry leaders.

## EDUCATION

**California State University, Fullerton**
M.S. Computer Science *August 2012*
B.S. Computer Science, Minor in Mathematics *July 2011*

## PUBLICATIONS AND PRESENTATIONS

C. Lee, H. Newman, Y. Wu, S. Axani. Measured WAN Traffic Reduction from Federated Edge AI in a Distributed Cosmic-Ray Sensor Network. INDIS 2026 workshop, SC26 Supercomputing Conference, Chicago, IL, 2026 (to appear).

C. Lee. Onboard AI and Federated Learning for Cosmic Ray Event Detection. Network Research Exhibition demonstration, SC25 Supercomputing Conference, St. Louis, MO, 2025.

K. Cheung, V. Vilnrotter, M. Sanchez-Net, C. Lee. High Data Rates from the Outer Solar System. IEEE Aerospace Conference, Big Sky, MT, 2025.

A. Agha et al. (72 authors), including C. Lee. NeBula: TEAM CoSTAR's Robotic Autonomy Solution that Won Phase II of DARPA Subterranean Challenge. Field Robotics, vol. 2, pp. 1432–1506, 2022.

J. Vander Hook, W. Seto, V. Nguyen, Z. Hasnain, C. Lee, L. Gallagher, T. Halpin-Chan, V. Varahamurthy, M. Angulo. Swarms of Pirates: Red Team Exercises Using Autonomous High-Speed Maneuvering Surface Vessels. Field Robotics, vol. 2, pp. 872–909, 2022.

C. Lee, M. Shaikh, C.H. Lee, D. Michels. Lunar Terrain Coverage Analysis Data Delivery Workflow. AIAA ASCEND 2021 (AIAA 2021-4039).

C. Lee, H. Xie, C.H. Lee, D. Lyakhov, D. Michels. In Silico Methods for Space System Analysis: Optical Link Coding Performance and Lunar Terrain Masks. AIAA ASCEND 2020.

D. Abraham, B. MacNeal, D. Heckman, Y. Chen, J. Wu, K. Tran, A. Kwok, C. Lee. Recommendations Emerging from an Analysis of NASA's Deep Space Communications Capacity. SpaceOps 2018, Marseille, France.

J. Lad, M. Johnston, D. Tran, D. Brown, K. Roffo, C. Lee. Complexity-Based Link Assignment for NASA's Deep Space Network for Follow-the-Sun Operations. SpaceOps 2018, Marseille, France.

K. Pinover, M. Johnston, C. Lee. Optimizing SmallSat Scheduling for NASA's Deep Space Network. International Workshop on Planning and Scheduling for Space (IWPSS), Pittsburgh, PA, 2017.

D. Morabito, D. Kahan, K. Oudrhiri, C. Lee. Cassini Downlink Ka-Band Carrier Signal Analysis. The Interplanetary Network Progress Report, vol. 42-208, 2017.

K. Cheung, D. Abraham, M. Sanchez-Net, K. Tran, C. Lee. Traffic Modeling for Deep Space Network in the Human Exploration Era. SpaceOps 2016, Daejeon, Korea.

M. Johnston, B. Carruth, M. Wallace, A. Coffman, C. Lee, C. Lau, K. Cheung, M. Levesque. Integrating Space Communication Network Capabilities via Web Portal Technologies. SpaceOps 2014, Pasadena, CA.

C. Lee, C.H. Lee. Cancer Screening Using Multi-modal Differential Principal Orthogonal Decomposition. International Conference on Computational Science and Its Applications (ICCSA), Ho Chi Minh City, Vietnam, pp. 39–45, 2013.

C. Lee, C.H. Lee. Extended Principal Orthogonal Decomposition Method for Cancer Screening. International Journal of Bioscience, Biochemistry and Bioinformatics, vol. 2, no. 2, pp. 136–141, 2012.

C. Lee. REST Architecture for Link Analysis Tools Portal. NASA Undergraduate Student Research Program (USRP), Pasadena, CA, 2011.